GEOG 4/5/7 9073: Environmental Analysis in R

Week 01.02: A quick and limited introduction to R

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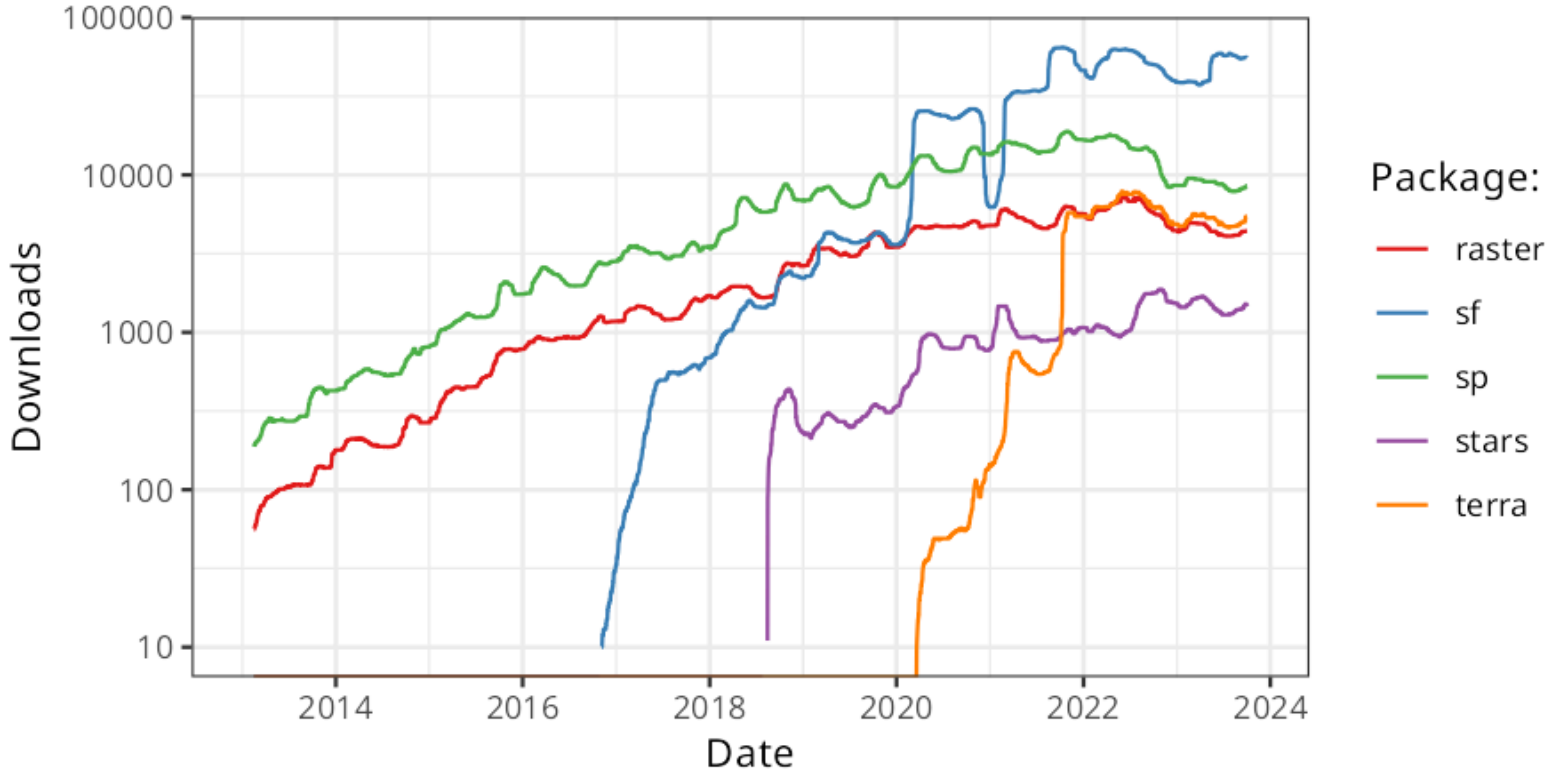
Today's schedule

- Open discussion
- R basics and practice

Anything to discuss? Questions?

A bit of background



R for Geospatial Analysis – Spatial Ecosystem (for more)

Let's review some GIScience

- Wait, what's GIScience?
- And how's it different than GIS?
- How do GIS and GIScience differ from Spatial Analysis?

A quick review

- Components of a GIS?
- Data types?
- Spatial functions?

Let's get started with some R

1. Open RStudio
2. Create a new project in a temporary working directory (it can be anywhere)
   a. File -> new Project
   b. What do you see?
   c. what's the `>`?
3. Writing in the console vs. writing code in a script
4. Let's talk about packages... (install **tidyverse**)
5. Assignment: "<-" vs. "="
6. ...and familiarize ourselves with the interface

A quick tour of the R environment

Some simple work

type the following in the console

```
x <- 7
```

- the `<-` is the "assignment operator"
- `=` also works, but can be incorrect in rare occasions
- so use `<-`

Did anything else change?

So we have a numeric value stored as a variable

Let's do stuff with it

Try the following

```
x + 2
x * 8
x / 1
x ** 2 # what does this do? <- and what is this ancillary text?
```

Let's try some "tests"

```
# what kind of tests are these? And what are they doing?
x / 2 == 0
x > 2
x == 7
```

Working with vectors

- A vector is a 1-D ordered collection of values
- you designate a vector in R with `c()`

For example, `y <- c(1, 2, 3, 4, 5)`

try it!

What does your environment viewer tell you?

R operations are "vectorized"

this means you can do things like this:

STOP: Before you execute this line... what do you think is going to happen?

```
y + 2
```

or you can do this:

```
c(1,2,3) + c(4,5,6)
```

But a vector can't mix data types

What do you think will happen if we try:

```
z <- c(1, 2, "3")
```

what DID happen?

Much of R is built on vectors, but a lot is also built on lists

Lists can mix and match data types

```
mylist <- list(6, 41, "banana")
```

Vectorized operations don't work on lists...

Accessing elements from a vector or a list

```
mylist[2]
```

what does this return?

and what does it tell you about R data structures?

Data frames

- A 2-dimensional data structure that has a lot in common with a common "table"
- Functionally, it's a list of lists

Let's break this down first. What do we expect to happen? How does the syntax work?

```
mydf <- data.frame(names = c("Huey", "Dewey", "Louis"),
  height = c(45, 42, 40))

#then...

mydf
```

What do you see in the console?

Packages

- Collections of code, function, and data written by others
- The foundation of the R ecosystem
- ONLY need to be "installed" ONCE
- Then need to read into memory for each session
- Packages of packages are a thing

```
#install it
install.packages("tidyverse") # Quotes here

#load it into memory
library(tidyverse) # no quotes here
```

Calling a function

once it's in memory, you can call a function from that package directly

```
filter(mydf, height > 43)
```

but namespace conflicts happen, so you can be explicit too

```
dplyr::filter(mydf, height > 43)
```

Getting help on a function

```
?dplyr::filter
```

or on a package

```
?dplyr
```

Try it, what happens?

Writing your own function

the syntax is a bit weird, so let's break it down

```
myfirstfunction <- function(x, y){
  x + y
}
```

then call the function (make sure it's in memory first)

```
myfirstfunction(4, 8)
```

If there's time...

- In small groups, figure out how you'd do the following:
- Write a function that takes two integers. If **both are even** or **both are odd**, the function returns **TRUE**. Otherwise, it returns **FALSE**

- Start with the algorithm, NOT the code
- Then try to write the function

Review and next class

- Any questions?

For next class (also on Canvas page)

1. First, download and install the R computing environment (just get the most recent one): https://www.r-project.org
2. Next, install RStudio Desktop, the IDE we'll use this semester. It's a user-friendly (well, more friendly at least) way to create projects https://www.rstudio.com/products/rstudio/download/

More next class (also on Canvas page)

3. If you don't already have a GitHub account, sign up for one. https://github.com
4. Then, get your GitHub Student Developer Pack. It grants you a LOT of FREE stuff: https://education.github.com/pack/join
5. We'll talk about git and GitHub in class, but there are many ways to use it. The command line is popular and powerful, but there is a steep learning curve. There are many free and paid software options as well. GitHub Desktop is one of them https://desktop.github.com, but feel free to use your Google-Fu and find something that you like

Homework for next week (also on Canvas page)

1. Complete the above setup steps
2. Review chapters 1 - 4 on the R for Data Science page I linked above. Don't worry, it'll go more quickly than it sounds
3. Come to Tuesday's class with **at least 2 questions** you have about R, geospatial programming in general, or this course.

- Next week’s readings/tasks:
  - Chapter 2 in textbook
  - Review Hadley's book/site
  - Practice on your own
- Next week's topics: data structures, data munging, plots 101